

GOVERNMENT DATA SOURCES

Pesticides, Soil, Food & Human Exposure

A research-ready source map for investigating whether agricultural chemical intensity and non-regenerative practices are associated with soil degradation, food nutrient or contaminant profiles, and human exposure.

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Central principle

Prefer direct laboratory measurements and mandatory administrative reporting. Use survey estimates and modeled datasets as supporting evidence, exposure proxies, or tools for hypothesis generation - not as exact ground truth.

Scope

This guide inventories official U.S. federal and California sources relevant to pesticide use, agricultural practice, soil condition, water contamination, food composition and residues, human biomonitoring, toxicology, incidents, and regulatory decision-making. It summarizes what each source can support, what it cannot, and how the pieces can be joined.

Policy jurisdiction

EPA generally registers, restricts, or cancels pesticides and sets tolerances under federal pesticide and food law. FDA monitors residues in FDA-regulated foods and enforces EPA tolerances. USDA holds most national farming-practice and soil-management data. An eventual campaign would therefore likely target EPA + USDA + FDA rather than FDA alone.

1. Executive source map

The federal government has collected nearly every part of the evidence chain, but the records are distributed across agencies and differ sharply in evidentiary strength.

Evidence layer	Strongest sources	Observation type	Primary question
Agricultural practices	Census of Agriculture; NRI; DSP4SH	Census/survey + field sampling	Where are cover crops, no-till, conventional tillage, and conservation practices used?
Pesticide application	California DPR PUR; NASS Chemical Use; USGS EPest	Mandatory reports; survey; model	Which chemicals were applied, where, when, and at what intensity?
Soil condition	RaCA; DSP4SH; SSURGO; NRI	Samples + mapped properties + estimates	What is the condition of soil carbon, structure, fertility, water capacity, and erosion?
Water	USGS Water Data; pesticide water-quality studies	Environmental samples	Which pesticides and metabolites occur in surface water or groundwater?
Food	FDA Residue Monitoring; FDA Total Diet Study; FoodData Central	Laboratory measurements	What residues, toxic elements, and nutrients are measured in foods?
Humans + toxicity	NHANES; CompTox; ToxCast; ToxRef/ToxVal	Biomonitoring + experimental/compiled evidence	What reaches people, and what biological effects are plausible at what doses?
Regulation + incidents	EPA registration dockets; Incident Data System	Regulatory record + passive surveillance	What did regulators know, decide, require, or receive as adverse reports?

A defensible causal claim requires more than a chain of correlations

- Separate inherent soil characteristics from management-sensitive soil health.
- Treat national pesticide-use estimates as exposure proxies unless validated against administrative records.
- Control for crop variety, geography, climate, irrigation, harvest maturity, storage, preparation, and analytical-method changes when studying food nutrients.
- Use chemical identifiers consistently: CASRN, EPA PC Code, DTXSID, active ingredient, and metabolite names.
- Pre-register key hypotheses and retain source versions, codebooks, and method documents.

Recommended framing

Across comparable crops and places, are pesticide intensity, conventional tillage, and absence of cover cropping associated with poorer measured soil-health indicators; and are those soil conditions associated with systematically different food nutrient or contaminant profiles?

2. Evidence hierarchy and capture-risk lens

Modeled or imputed data are not automatically captured or unusable. The key questions are whether underlying observations are inspectable, whether assumptions are documented, whether uncertainty is quantified, and whether results can be validated against independent measurements.

Tier	Type	Examples	How to use it	Main vulnerability
1	Direct physical measurement	FDA labs, RaCA, USGS water, NHANES	Primary evidence, subject to sampling design and laboratory limits	Non-random sampling, detection limits, method drift
2	Mandatory administrative report	California DPR PUR	Primary application evidence; aggregate carefully	Misreporting, enforcement gaps, confidential detail
3	Census or probability survey	Census of Agriculture, NASS Chemical Use, NRI	Population estimates with uncertainty and methodology	Nonresponse, weighting, imputation, rotation gaps
4	Modeled estimate	USGS EPest, some SSURGO interpretations	Screening, spatial prioritization, sensitivity analyses	False geographic precision; model dependence
5	Compiled regulatory or surveillance record	EPA dockets, CompTox, incidents	Institutional knowledge, hazards, decisions, signal detection	Selective reporting, heterogeneous studies, inaccessible inputs

The national pesticide-use transparency problem

Outside California, USGS county estimates combine harvested acreage with surveyed pesticide-use rates and geographic allocation. Missing information can require extrapolation, and USGS publishes low/high estimates for alternative treatments of missing data. EPA also licenses nationwide agricultural pesticide-use information from a commercial provider that cannot be redistributed as raw public records. This creates an epistemic dependency: the public may be unable to reconstruct some nationally used exposure inputs from inspectable observations.

That situation is evidence of a transparency and reproducibility problem. It is not, by itself, proof of regulatory capture. Capture claims require additional evidence about institutional incentives, conflicts, rulemaking, suppressed evidence, revolving-door relationships, enforcement, or systematic divergence between independent measurements and regulatory conclusions.

Audit questions to record for every dataset

- Who generated the observations, who funded collection, and who owns the raw data?
- Is reporting mandatory, voluntary, sampled, or inferred?
- Are row-level data, codebooks, sampling frames, response rates, and uncertainty available?
- Which values are measured, imputed, suppressed, or modeled?
- Can a second source validate the direction and magnitude of the result?

3. Source catalog: pesticide use and farming practices

1. California Pesticide Use Reporting (PUR)

REPORTED

Agency	California Department of Pesticide Regulation	Priority	Highest
Contains	Reported pesticide applications with chemical/product, amount, crop or site, date, and location fields; annual summary and downloadable data products.	Access	CDPR pesticide-use data
Best use	Primary application evidence and validation benchmark for national estimates; ideal first geography for a natural-laboratory design.	Cautions	Reported is not identical to observed. Audit reporting requirements, revisions, location precision, and confidentiality/suppression rules.

2. Agricultural Chemical Use Program

SURVEY

Agency	USDA National Agricultural Statistics Service	Priority	High
Contains	Commodity-by-state survey statistics: active ingredients, acres treated, percent treated, total applied, application rates, fertilizers, and pest-management practices on rotating crop schedules.	Access	Program, data, and methods
Best use	National/state estimates of chemical intensity by crop; triangulation with PUR and residue results.	Cautions	Probability survey with weighting, nonresponse, imputation, rotating commodity/state coverage, and confidentiality suppression. Not annual universal reporting.

3. Estimated Agricultural Pesticide Use (EPest)

MODELED

Agency	U.S. Geological Survey	Priority	Supporting
Contains	Annual county-level low/high estimates by pesticide compound, based on proprietary use surveys, harvested acreage, and allocation procedures; historical national spatial coverage through 2019 releases.	Access	USGS county pesticide-use release
Best use	Exposure proxy, hotspot screening, spatial prioritization, and national sensitivity analysis.	Cautions	Do not state exact county application as fact. Underlying commercial survey microdata are not publicly reproducible; missing observations and downscaling can create apparent precision.

4. Census of Agriculture

SURVEY

Agency	USDA National Agricultural Statistics Service	Priority	Highest
Contains	County/state counts and acres for no-till, reduced/conservation tillage, intensive/conventional tillage, cover crops, irrigation, farm type, production, and operator characteristics.	Access	2022 Census data
Best use	Construct county practice shares and stratify or match farming systems.	Cautions	Five-year cadence; operator-reported practices; disclosure suppression; changing question wording and universe definitions.

5. National Resources Inventory (NRI)

SURVEY

Agency	USDA Natural Resources Conservation Service	Priority	High
Contains	Longitudinal statistical inventory of non-federal land: land use, conservation practices, irrigation, wetlands, and wind/sheet/rill erosion estimates.	Access	NRI data and documentation
Best use	National/state trend estimates and erosion outcomes, including tons of soil loss per acre per year.	Cautions	Designed for aggregate inference; exact sample locations are restricted and public outputs may not support fine-grained joins.

4. Source catalog: soil condition and water

6. Rapid Carbon Assessment (RaCA)

MEASURED

Agency	USDA Natural Resources Conservation Service	Priority	Highest
Contains	Approximately 145,000 samples from more than 32,000 profiles at roughly 6,000 locations, with organic/inorganic carbon, bulk density, depth, land cover, and related site data.	Access	RaCA downloads and documentation
Best use	Measured soil-carbon stocks and landscape baselines; test differences by land cover and selected management context.	Cautions	Primarily a national baseline campaign rather than a dense repeated time series. Sampling support and spatial representativeness must be respected.

7. Dynamic Soil Properties for Soil Health (DSP4SH)

MEASURED

Agency	USDA Natural Resources Conservation Service	Priority	Highest / emerging
Contains	Management-sensitive field measurements including soil organic carbon, bulk density, aggregate stability, soil protein, permanganate-oxidizable carbon, and enzyme activity across business-as-usual, soil-health, and perennial references.	Access	DSP4SH program
Best use	Closest federal source to testing whether regenerative/soil-health management produces measurable soil differences.	Cautions	Confirm public availability, sampling coverage, release version, management definitions, and comparability before designing a national analysis.

8. SSURGO / Soil Data Access

MODELED

Agency	USDA Natural Resources Conservation Service	Priority	High as covariate
Contains	Detailed mapped soil components and horizons: organic matter/carbon-related fields, pH, texture, cation exchange capacity, bulk density, water capacity, salinity, root-zone and erosion interpretations.	Access	SSURGO access
Best use	Baseline soil geography, covariates, stratification, and spatial joins to cropland or watersheds.	Cautions	Many values represent mapped, representative, or interpreted properties rather than current repeated soil-health measurements. Avoid treating every polygon attribute as a contemporaneous sample.

9. Water Data for the Nation + pesticide studies

MEASURED

Agency	U.S. Geological Survey	Priority	Highest
Contains	Site-based surface-water and groundwater observations, laboratory parameters, streamflow context, and pesticide occurrence studies and data releases.	Access	Water Data for the Nation Pesticides and water quality
Best use	Environmental occurrence and transport; watershed or monitoring-site comparisons near agricultural intensity gradients.	Cautions	Sampling is uneven in space/time and often purpose-driven. Non-detection depends on analyte list and reporting limit; site attribution is not farm attribution.

5. Source catalog: food composition and residues

10. Pesticide Residue Monitoring Program

MEASURED

Agency	U.S. Food and Drug Administration	Priority	Highest
Contains	Sample-level analytical records for domestic and imported FDA-regulated foods, including commodity, origin, pesticide detections, concentration/results, and compliance context; annual reports and downloadable files.	Access	FDA reports and datasets
Best use	Determine which residues are detected in foods, compare commodities/origins/years, and link chemicals to toxicology records.	Cautions	Compliance-oriented sampling is not necessarily representative of the food supply. Non-detection depends on the analytical scope and limits; changing methods complicate trends.

11. Total Diet Study (TDS)

MEASURED

Agency	U.S. Food and Drug Administration	Priority	Highest
Contains	Foods purchased and prepared as eaten, assayed for nutrient elements, toxic elements, pesticide residues, industrial chemicals, and radionuclides; historical and modernized releases.	Access	FDA Total Diet Study
Best use	Core source for micronutrients and contaminants in prepared foods; estimate dietary exposure using consumption models.	Cautions	Food lists, sampling design, preparation, laboratory methods, and detection limits changed across eras. Long-term trends require harmonization and method-break sensitivity tests.

12. FoodData Central

MEASURED

Agency	USDA Agricultural Research Service	Priority	High
Contains	Foundation Foods analytical records, historical Standard Reference, FNDDS survey foods, experimental foods, and branded-label records in CSV/JSON plus API access.	Access	Downloadable datasets API guide
Best use	Independent nutrient composition source; use Foundation Foods and underlying analytical samples where possible.	Cautions	Do not mix analytical and label-derived data without source-type controls. Foods, cultivars, fortification, moisture basis, and sample aggregation change over time.

Critical warning for the micronutrient hypothesis

A decline in a nutrient value across database releases is not automatically a decline in crops. It may reflect different cultivars, growing regions, maturity, edible portions, moisture, fortification, recipes, sampling frames, laboratory methods, or database aggregation. The strongest design follows matched crops/varieties and documents every method transition.

6. Source catalog: human exposure, toxicology, incidents, and decisions

13. NHANES Laboratory Data

MEASURED

Agency	CDC National Center for Health Statistics	Priority	High
Contains	Blood and urinary biomarkers for selected pesticide metabolites and environmental chemicals, joined to demographic, dietary, occupational, and health variables through participant identifiers.	Access	NHANES laboratory catalog
Best use	Population exposure trends and disparities; biomarker associations with health or occupational factors using survey weights.	Cautions	Cross-sectional design, short biological half-lives for many metabolites, changing analyte panels, values below detection, and complex survey design. Cannot identify a source farm.

14. CompTox Chemicals Dashboard / CTX APIs

REGULATOR
Y

Agency	U.S. Environmental Protection Agency	Priority	Highest for linkage
Contains	Chemical identities, physicochemical properties, fate, exposure, in vivo toxicity, in vitro bioactivity, hazard values, and links across more than one million chemicals.	Access	CompTox Dashboard CTX APIs
Best use	Resolve chemical identifiers and join pesticide-use/residue records to toxicity, exposure, and regulatory endpoints.	Cautions	A compilation of heterogeneous sources and models. Trace decisive values to original studies and preserve provenance; API access may require a free key.

15. ToxCast / invitrodb

MEASURED

Agency	U.S. Environmental Protection Agency	Priority	High
Contains	High-throughput concentration-response assays across biological targets and pathways; downloadable database, assay annotations, quality-control outputs, and reproducible R tooling.	Access	ToxCast downloads
Best use	Hazard screening, mechanism hypotheses, chemical prioritization, and comparison of bioactivity potency across compounds.	Cautions	In vitro bioactivity is not equivalent to in vivo harm at real-world exposure. Cytotoxicity, assay interference, pharmacokinetics, and exposure context must be considered.

16. Pesticide Incident Data System (IDS)

REGULATOR
Y

Agency	U.S. Environmental Protection Agency	Priority	Supporting
Contains	Reported adverse incidents involving humans, animals, wildlife, plants, and the environment, with product/ingredient, severity, location, and sometimes narratives.	Access	EPA Incident Data System
Best use	Signal detection, case characterization, and identification of recurring products, settings, or harms.	Cautions	Passive surveillance: underreporting, stimulated reporting, duplicates, incomplete narratives, uncertain causality, and denominator absence.

17. Pesticide Registration Review Dockets

REGULATOR
Y

Agency	U.S. Environmental Protection Agency / Regulations.gov	Priority	Highest for accountability
Contains	Preliminary/final work plans, use summaries, data call-ins, human-health and ecological risk assessments, public comments, proposed/interim/final decisions, and mitigation actions.	Access	EPA registration review process
Best use	Compare independent findings with what EPA considered, requested, concluded, and regulated; identify open comment windows and evidentiary gaps.	Cautions	A decision record, not an unbiased dataset. Evidence may be registrant-submitted; documents and assumptions require close review, and docket completeness varies.

Regulatory accountability workflow

For each priority chemical, build a compact evidence dossier that compares measured observations with EPA's stated assumptions and decisions.

Step	Extract	Question
1. Scope	Registered uses, crops, rates, exposure pathways	Does the review cover the uses visible in application data?
2. Inputs	Submitted studies, monitoring data, usage estimates, models	Which decisive inputs are public and independently reproducible?
3. Assumptions	Exposure scenarios, uncertainty factors, endpoints	Are assumptions consistent with measured residues, water, or biomarkers?
4. Decision	Risk findings, mitigations, tolerances, deadlines	What evidence threshold triggered - or failed to trigger - action?
5. Challenge	Contradictions, gaps, newer evidence, comment windows	What precise data request or regulatory remedy follows?

A strong petition cites the agency's own docket language, identifies the exact assumption contradicted by public measurements, supplies reproducible analysis, and requests a remedy within the agency's legal authority.

7. How the datasets connect

There is no universal key. Build explicit crosswalks and analyze at the coarsest geography and time unit supported by all joined sources.

Join	Likely keys	Use	Risk
Application → soil	Year; county/FIPS; coordinates; crop/site; watershed	Relate reported or estimated intensity to soil outcomes	Soil sampling may predate/follow exposure; ecological fallacy
Practice → soil	County/state; crop; land-use class; sampling stratum	Compare cover crop/tillage prevalence with management-sensitive measures	County shares do not identify management at a sample point
Application → water	Watershed/HUC; monitoring site; upstream cropland; date/season	Test occurrence and transport following application windows	Hydrology, lag, degradation products, and mixed sources
Chemical → residue/toxicity	CASRN; DTXSID; EPA PC Code; standardized ingredient/metabolite	Connect detected residues to hazard and regulatory records	Parent compounds, salts, isomers, degradates, mixtures
Food → nutrient	Commodity; food code; cultivar; preparation; year; origin	Study nutrient/contaminant profiles	Food descriptions and laboratory methods are not stable
Exposure → population	NHANES participant ID; survey cycle; weights; demographics	Estimate national exposure distributions and disparities	Cannot directly join individual biomarkers to farms or food samples

Minimum provenance table

Field	Record	Why
Dataset/version	Title, release date, retrieval date, URL, file checksum	Makes the analysis reproducible
Observation basis	Measured, reported, survey, imputed, modeled, compiled	Prevents evidence-tier confusion
Unit/coverage	Record unit, geography, time, population/crop universe	Defines valid inference
Missingness	Suppression codes, nondetects, nonresponse, imputation flags	Missingness may be structurally informative
Method	Sampling frame, lab method, reporting limit, model version	Needed for trend and cross-source comparisons
Identifiers	FIPS/HUC, food codes, CASRN/DTXSID/PC Code	Prevents false joins

Never discard the low/high EPest pair, nondetect flags, survey weights, imputation codes, or method version. Those are analysis variables, not housekeeping.

8. Recommended collection sequence

Start with a California pilot, where application records are materially closer to auditable primary evidence. Only scale nationally after the pipeline and hypotheses survive that test.

Phase	Acquire	Output	Decision gate
1. Provenance	Codebooks, methods, release notes, coverage tables	Source register + data dictionary	Can measured/reported/imputed values be distinguished?
2. California exposure	CDPR PUR + crop/land geography	Chemical × crop × place × time intensity table	Are reporting units and locations sufficiently stable?
3. Soil + water	RaCA/DSP4SH where available; SSURGO covariates; USGS water	Measured outcomes + baseline covariates	Is overlap adequate for credible comparisons?
4. Food	FDA Residue Monitoring + TDS + Foundation Foods	Commodity residue/nutrient panels	Can foods, methods, and origins be harmonized?
5. Hazard + regulation	CompTox/ToxCast + EPA dockets + incidents	Chemical evidence dossiers	Do measured signals conflict with regulatory assumptions?
6. National comparison	NASS Chemical Use + EPest + Census + NRI + NHANES	Sensitivity and external-validity analysis	Do results persist across model choices and uncertainty?

First six downloads

- California DPR PUR - mandatory application records.
- FDA Pesticide Residue Monitoring - sample-level food detections.
- FDA Total Diet Study - nutrients, toxic elements, and pesticide results.
- USDA RaCA - measured carbon and bulk-density baseline.
- 2022 Census of Agriculture - tillage and cover-crop practice layer.
- USGS water-quality observations - measured environmental occurrence.

Then acquire NASS Chemical Use and USGS EPest to test national generalization, not to anchor the initial factual claim.

Early analytical outputs

- A source/provenance ledger with evidence tier and public reproducibility score.
- A chemical crosswalk linking active ingredients, metabolites, CASRN, DTXSID, and EPA PC Codes.
- A California application-intensity map with reporting completeness checks.
- A preregistered pilot comparing matched crop/place/time groups, with alternative exposure measures.
- A gap analysis naming which regulatory claims depend on inaccessible or highly modeled inputs.

9. Bottom line

The project is feasible, but the cleanest initial claim is narrower than 'non-regenerative farming strip-mines soil and reduces nutrients.' The first goal should be to establish a transparent, auditable chain of evidence and identify exactly where direct observation ends and modeling begins.

California is the strongest starting point because mandatory pesticide reporting can be evaluated alongside measured soil, water, food, and toxicology sources. National USGS pesticide estimates remain useful for screening and external-validity analysis, but should not be presented as precise county-level application totals without uncertainty and validation.

A powerful data-activism result could be substantive, procedural, or both: a demonstrated environmental/health association; a mismatch between independent measurements and regulatory assumptions; or a finding that crucial national pesticide-use inputs cannot be independently audited because the raw data are commercially controlled.

Official access points

Source	Official URL
USDA NASS Agricultural Chemical Use	https://www.nass.usda.gov/Surveys/Guide_to_NASS_Surveys/Chemical_Use/
USGS Estimated Agricultural Pesticide Use	https://www.usgs.gov/data/preliminary-estimated-annual-agricultural-pesticide-use-counties-conterminous-united-states-0
California DPR Pesticide Use	https://www.cdpr.ca.gov/pesticide-use-in-california/
FDA Pesticide Residue Monitoring	https://www.fda.gov/food/pesticides/pesticide-residue-monitoring-program-reports-and-data
FDA Total Diet Study	https://www.fda.gov/food/reference-databases-and-monitoring-programs-food/fda-total-diet-study-tds
USDA NRCS RaCA	https://www.nrcs.usda.gov/resources/data-and-reports/rapid-carbon-assessment-raca
USDA NRCS DSP4SH	https://www.nrcs.usda.gov/resources/education-and-teaching-materials/dynamic-soil-properties-for-soil-health-dsp4sh
USDA NRCS SSURGO	https://www.nrcs.usda.gov/resources/data-and-reports/soil-survey-geographic-database-ssurgo
USDA NRCS NRI	https://www.nrcs.usda.gov/nri
2022 Census of Agriculture	https://data.nass.usda.gov/Publications/AqCensus/2022/
USDA FoodData Central downloads	https://fdc.nal.usda.gov/download-datasets/
CDC NHANES laboratory data	https://wwwn.cdc.gov/nchs/nhanes/search/datapage.aspx?Component=Laboratory
EPA CompTox Dashboard	https://www.epa.gov/comptox-tools/comptox-chemicals-dashboard
EPA ToxCast data	https://www.epa.gov/comptox-tools/exploring-toxcast-data
EPA CTX APIs	https://www.epa.gov/comptox-tools/computational-toxicology-and-exposure-apis
EPA Pesticide Registration Review	https://www.epa.gov/pesticide-reevaluation/registration-review-process
EPA Incident Data System	https://www.epa.gov/pesticide-incidents/about-incident-data-system-ids
USGS Water Data for the Nation	https://waterdata.usgs.gov/

Prepared from official agency documentation accessed 14 September 2026. Dataset release dates, schemas, and access mechanisms can change; retain local copies of documentation with every downloaded version.